

## **UNIT 2 — Node.js**

### **1. Getting Started with Node.js — Runtime, Architecture & Environment**

- Introduction to Node.js
- What is Node.js? Why Node.js?
- Node.js Architecture Overview
- Runtime Environment: JavaScript Outside the Browser
- How V8, libuv & the Node APIs Fit Together
- Features of the Node.js Runtime
- REPL
- Running Node.js and Writing Basic Scripts

### **2. Global Objects and Modules**

- Working with JavaScript in Node.js
- Introduction to Modules
- CommonJS modules: `require` / `module.exports`

### **3. Node.js Internals — V8 Engine & libuv Deep Dive, The Node.js Event Loop In-Depth, Asynchronous Programming — Callbacks, Promises & Async/Await**

- V8 Engine and JavaScript Compilation
- libuv And Cross-Platform Abstraction
- Introduction to Asynchronous Programming
- Understanding Callbacks
- Callbacks and Error-First Convention
- Under the Hood of Asynchronous Execution: Core components of asynchronous execution
- The Node.js Event Loop In Depth
- Using Promises : What is promise, Creating a promise, Chaining the Promise
- Promise combinators — `Promise.all`, `allSettled`, `race`, `any` (new topic)
- Async/Await Syntax

#### **4. Event-Driven Programming**

- Introduction to Event-Driven Programming -EventEmitter — on, emit, once, removeListener
- Managing Multiple Listeners
- Dynamic Event Handling

#### **5. Core Node.js Modules**

- Working with Files and Directories
- Working with Path, OS, FS, http and Util Modules
- Introduction to Buffers
- Working with Streams

#### **6. npm and Package Management**

- Introduction to npm
- Working with npm Packages (package.json & package-lock.json)
- Managing Dependencies (Dev vs production dependencies)
- Exploring Commonly Used npm Packages (lodash, dotenv, nodemon)
- Introduction to ES6 Modules in Node.js

#### **7. Building a Basic Web Server**

- HTTP Request Structure: Headers
- HTTP Methods (GET vs POST)
- HTTP Response Structure & HTTP Status Codes
- Creating a Simple HTTP Server: http Module — createServer, req, res
- Basic Routing by URL & Method
- Serving HTML and JSON Responses
- Handling GET and POST Parameters: Query Params & POST Body

## **8. Connecting Node.js to MongoDB**

- Connecting to MongoDB via Node.js
- Connecting and Performing CRUD Operations
- Exercise: Build a REST API for a `students` Collection — Full CRUD via Routes

## **9. Cookies, Sessions, Same-Origin Policy & Authentication Basics**

- Same-Origin Policy (SOP): Origin (Protocol, Domain, Port), Browser Restrictions
- What are Cookies? Setting & Reading Cookies
- Session Management Concepts
- Authentication Overview: Session-Based vs Token-Based Authentication
- Implementing Authentication in Node.js: JWT Basics
- Hashing Passwords with bcrypt (Intro)

## **10. CORS & Real-Time Communication with WebSockets**

- HTTP vs WebSocket Protocol
- ws Library: Web Server & Client Setup
- Broadcasting Messages
- Rooms & Namespaces Concept
- CORS: What It Is and Why It Matters
- CORS HTTP Headers; Simple Requests vs Preflight Requests